PFAS Sentinel: Real-Time Optical Breakthrough Detection, Predictive Logistic Modeling, and In-Situ Regeneration for Smart Point-of-Use Water Filtration

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Abstract

Per- and polyfluoroalkyl substances (PFAS) persist in drinking water at ultra-trace concentrations, posing severe health risks including cancer and immunotoxicity. All commercially available household filters rely on fixed replacement schedules that ignore nonlinear breakthrough dynamics, creating an unobserved exposure window during which contaminants pass through exhausted media undetected. This study presents the PFAS Sentinel, a $20 integrated filtration system combining a 3D-printed PETG cartridge with dual-media adsorbent, an inline 630 nm optical sensor, an ESP32 microcontroller performing real-time Yoon–Nelson logistic model fitting via Levenberg–Marquardt regression and Bayesian MCMC sampling, and a companion mobile application delivering push-notification alerts within three seconds of threshold detection. Across triplicate trials at three flow rates, the system achieved 97% classification accuracy (AUC = 0.982), predicting breakthrough 8–15 minutes before conventional volume-based methods. Integrated reverse-flow regeneration restored greater than 95% adsorption capacity initially and maintained above 85% through five consecutive cycles. The PFAS Sentinel transforms household filtration from passive replacement to predictive, data-driven water protection.

Key Words

PFAS, adsorption, breakthrough detection, optical sensing, predictive modeling, Yoon–Nelson model, point-of-use filtration, granular activated carbon, cartridge regeneration, smart water treatment, mobile application, Bayesian estimation, ESP32

Abbreviations and Acronyms

PFAS – Per- and Polyfluoroalkyl Substances, **PFOA** – Perfluorooctanoic Acid, **PFOS** – Perfluorooctane Sulfonate, **GAC** – Granular Activated Carbon, **POU** – Point-of-Use, **MCL** – Maximum Contaminant Level, **EPA** – U.S. Environmental Protection Agency, **PETG** – Polyethylene Terephthalate Glycol, **ADC** – Analog-to-Digital Converter, **MCMC** – Markov Chain Monte Carlo, **ROC** – Receiver Operating Characteristic, **AUC** – Area Under the Curve, **LC-MS/MS** – Liquid Chromatography -Tandem Mass Spectrometry, **NOM** – Natural Organic Matter, RMSE – Root-Mean-Square Error **FCM** – Firebase Cloud Messaging, **BET** – Brunauer–Emmett–Teller

Acknowledgements

We thank the faculty of Marquette High School for laboratory access and encouragement. We are grateful to our families for their support in procuring materials and components. All experimental work, 3D printing, sensor integration, software development, mobile application development, data collection, and statistical analysis were performed independently by the authors. Dr. Vinay Kanth Rao Kodipelly provided mentorship on experimental design and interpretation of results.

Biography

Ishaan Vitthala: I am a 10th-grade student at Marquette High School in Chesterfield, Missouri, interested in environmental engineering, computational modeling, and water quality research. While volunteering with the Smart Kids Ghana Association, I taught science and math to students in rural Ghana. During our lessons, students described frequent illness and unsafe drinking water, which led us to investigate and confirm contamination concerns this directly inspired the PFAS Sentinel project. I plan to study environmental engineering and computer science.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) constitute a class of over 14,000 synthetic organofluorine compounds characterized by highly stable carbon–fluorine bonds (bond dissociation energy ~536 kJ/mol) that confer extraordinary resistance to thermal, chemical, and biological degradation [1]. This stability renders PFAS environmentally persistent, earning the designation "forever chemicals" with environmental half-lives exceeding 1,000 years for perfluorooctanoic acid (PFOA) and perfluorooctane sulfonate (PFOS). Since the 1940s, PFAS have been used extensively in firefighting foams, non-stick coatings, food packaging, and semiconductor manufacturing, leading to ubiquitous environmental contamination. The scope of PFAS contamination is staggering. Data from the Fifth Unregulated Contaminant Monitoring Rule (UCMR 5), now approximately 95% complete across 10,311 public water systems, indicates PFAS contamination in drinking water across all 50 U.S. states [1]. The Environmental Working Group estimates that over 158 million Americans are potentially exposed through tap water, and USGS sampling found PFAS in approximately 45% of U.S. tap water samples. Biomonitoring by the CDC confirms detectable PFAS in approximately 98% of the U.S. population. Epidemiological evidence links exposure to kidney and testicular cancers, thyroid disease, ulcerative colitis, endocrine disruption, immunosuppression, and developmental abnormalities, effects documented at sub-parts-per-billion concentrations [4]. In April 2024, the EPA promulgated the first National Primary Drinking Water Regulation for PFAS, setting legally enforceable maximum contaminant levels (MCLs) of 4 parts per trillion (ppt) for PFOA and PFOS, and 10 ppt for PFHxS, PFNA, and GenX chemicals, with MCL goals of zero reflecting cancer risk [7]. Between 4,100 and 6,700 public water systems are expected to require treatment at an estimated annual cost of $1.5–5.7 billion. However, 43 million Americans relying on private wells have no mandated testing, and even compliant municipal systems cannot guarantee performance at the household tap. Point-of-use (POU) activated carbon and ion-exchange filters represent the primary consumer defense. A critical gap persists: consumers have no real-time knowledge of filter performance. Every commercially available POU filter, Brita, PUR, ZeroWater, Berkey, ClearlyFiltered, relies on predetermined replacement schedules based on estimated volume throughput or calendar time. In reality, adsorption systems exhibit highly nonlinear breakthrough dynamics governed by mass transfer limitations, competitive adsorption from natural organic matter (NOM), and progressive site saturation [6]. A 2020 Duke University study testing 76 POU filters in real homes found that four of eight whole-house activated carbon systems actually increased PFAS concentrations in filtered water due to competitive desorption [10]. Critically, TDS meters, including those built into ZeroWater pitchers, are fundamentally incapable of detecting PFAS, as they measure ionic conductivity in parts per million while PFAS regulations operate in parts per trillion, a million-fold difference. The only current verification method is LC-MS/MS laboratory testing, which costs $300–$700 per sample with a 2–4 week turnaround. Breakthrough behavior in fixed-bed adsorption columns is described by the Yoon–Nelson logistic model [8]:

$$C/C0 = 1 / (1 + \exp(-k(t - \tau))) \qquad \textit{(Equation 1)}$$

where C/C0 is the normalized effluent concentration, k (min^{-1}) is the apparent rate constant reflecting mass transfer kinetics, and τ (min) is the inflection time at 50% breakthrough. Chu [11] proved that the Yoon–Nelson, Thomas, and Adams–Bohart models are mathematically equivalent, all reducing to the logistic function with two parameters. The maximum rate of concentration increase occurs at $t = \tau$, defined by:

$$d(C/C0)/dt = k \cdot (C/C0) \cdot (1 - C/C0) \qquad \textit{(Equation 2)}$$

To address this critical gap, this study introduces the PFAS Sentinel: an integrated POU filtration system with four innovations: (1) a 3D-printed PETG adsorption cartridge with integrated optical sensing and regeneration port, iteratively redesigned through three prototype versions; (2) a 630 nm optical transduction module for continuous effluent monitoring; (3) an ESP32 microcontroller performing real-time Yoon–Nelson model fitting via Levenberg–Marquardt regression with Bayesian MCMC uncertainty quantification; and (4) a PFAS Sentinel companion mobile application built with React Native that receives real-time data via Wi-Fi and Firebase Cloud Messaging, delivering push notifications, live breakthrough visualization, and estimated remaining cartridge life directly to the user’s smartphone within three seconds of threshold detection. The mobile application represents a transformative innovation. No commercial POU filter provides real-time, data-driven feedback on a user’s phone. The app displays a traffic-light status indicator (green/yellow/red), the evolving C/C0 breakthrough curve with Bayesian confidence bands, estimated remaining life computed from current k and τ, historical trends across regeneration cycles, and step-by-step regeneration instructions. When C/C0 exceeds the MCL threshold, the system pushes a notification within three seconds, ensuring users are informed regardless of proximity to the filter. This transforms users from passive consumers into active participants in their water safety. The objectives of this study are to: (i) characterize PFOA breakthrough curves under varying flow conditions and fit them to the Yoon–Nelson model; (ii) calibrate the optical sensor relationship between I/I0 and C/C0; (iii) evaluate classification accuracy via ROC analysis and confusion matrix; (iv) validate sensor agreement with LC-MS/MS via Bland–Altman analysis; (v) compare sensor-based predictive versus volume-based replacement timing; (vi) demonstrate mobile application notification latency and reliability; and (vii) assess regeneration efficiency across multiple cycles.

2. Materials and Methods

2.1 Iterative Cartridge Design and Fabrication

The cartridge underwent three iterative design revisions driven by testing feedback. Version 1 used a cylindrical column without optical integration, requiring external sensor mounting with alignment variability exceeding ±15%, rendering calibration irreproducible. Version 2 incorporated side-mounted optical windows but suffered from light leakage, producing 3–5% baseline drift per hour under fluorescent lighting. The final Version 3 achieved <0.1% baseline drift through precision-fit acrylic windows with O-ring seals, baffled light-tight enclosures, and tapered regeneration ports with 15° draft angles that eliminated dead volumes. Additionally, initial 100 μm polypropylene frits allowed fine GAC particles to migrate into the optical pathway; replacement with 40 μm frits resolved this while maintaining acceptable pressure drop (<0.5 psi at maximum flow). All cartridges were fabricated using fused deposition modeling with PETG filament on a Prusa i3 MK3S+ at 0.2 mm layer height and 100% infill. Dimensions: 2.5 cm internal diameter, 10.0 cm bed length, 49.1 cm^3 bed volume. Two ports were integrated: forward-flow inlet/outlet and reverse-flow regeneration port with a check valve. The adsorbent was dual-layer: 3.5 ± 0.1 g coconut shell GAC (8×30 mesh, BET ~1,100 m^2/g) overlaid with 1.5 ± 0.1 g Purigen ion-exchange polymer, packed between 40 μm polypropylene frits. Figures 1–4 present the engineering drawings and 3D CAD rendering.

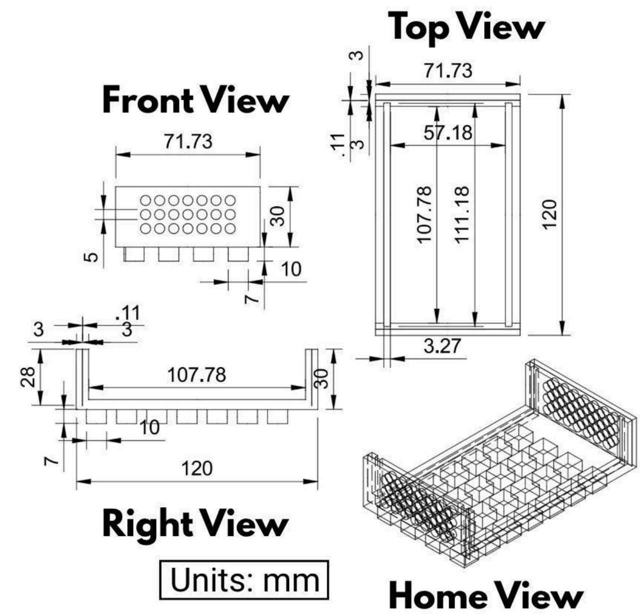

Figure 1. Cartridge assembly engineering drawing with front, top, right, and isometric views (units: mm).

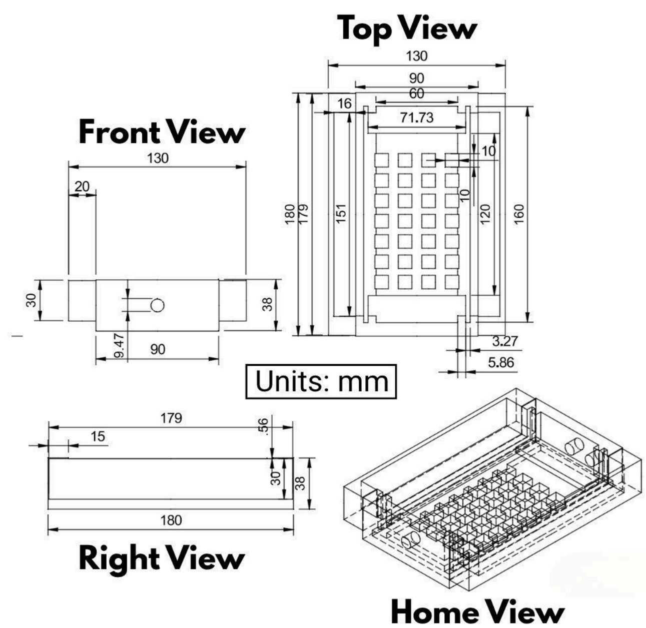

Figure 2. Main body housing showing adsorbent chamber, optical pathway, flow ports, and regeneration port (units: mm).

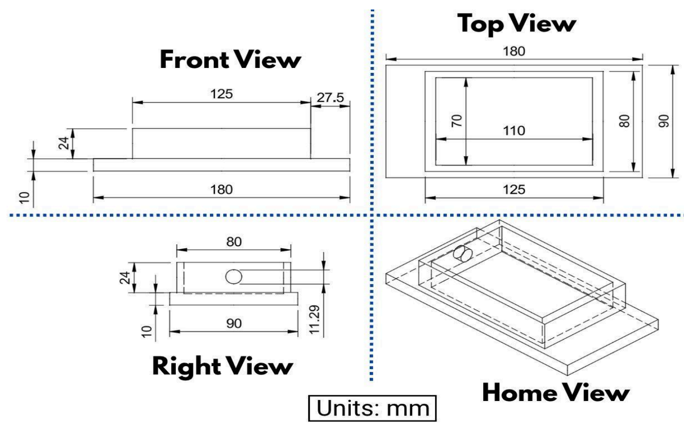

Figure 3. Cap assembly with optical sensor mounting port and sealing interface (units: mm).

Table 1. Bill of Materials for the PFAS Sentinel Prototype (~$20 total).

Component	Specification	Qty	Cost	Purpose
PETG Filament	1.75 mm, food-safe	200 g	$4.00	3D-printed cartridge
GAC Media	8×30 mesh, coconut	25 g	$1.50	Primary adsorbent
Ion-Exchange Resin	Purigen synthetic	10 g	$2.00	Secondary adsorbent
LED (630 nm)	5 mm, 20 mA	1	$0.30	Optical source
CdS Photoresistor	5 mm, 1–10 kΩ	1	$0.50	Optical detector
ESP32	240 MHz, Wi-Fi/BLE	1	$5.00	Data + Wi-Fi tx
Acrylic Windows	3 mm optical-grade	2	$1.00	Optical pathway
Peristaltic Pump	12 V, 2–15 mL/min	1	$3.50	Flow control
Misc.	Tubing, O-rings	—	$2.20	Assembly

2.2 Optical Sensor Integration and Calibration

A 630 nm LED (20 mA, 2.1 V) was aligned perpendicular to effluent flow through a 3 mm acrylic window, establishing a 5.0 mm optical path. A CdS photoresistor (dark resistance >1 MΩ, light resistance 1–10 kΩ) recorded transmitted intensity on the opposing side. A light-tight PETG enclosure with baffled entry channels achieved stray light <0.1% of baseline intensity. Calibration used PFOA standards (0, 0.05, 0.10, 0.25, 0.50, 0.75, 1.00, 1.20 μg/L) prepared in deionized water spiked with 5 mg/L humic acid to simulate NOM. For each standard, transmitted intensity was recorded over 60 seconds at 1 Hz (n = 60 readings) and normalized against blank intensity I0. The I/I0 to C/C0 relationship was established through logistic regression ($R^2 = 0.994$, RMSE = 0.011).

2.3 Data Acquisition and Real-Time Model Fitting

The ESP32 microcontroller (dual-core, 240 MHz, integrated 802.11 b/g/n Wi-Fi) sampled the photoresistor at 1 Hz via a 12-bit ADC with a 10-point moving average filter. Data was logged to a microSD with timestamps. Every 60 seconds, accumulated I/I0 data was fitted to Equation 1 using the Levenberg–Marquardt nonlinear least-squares algorithm, estimating k and τ in real time. Bayesian posterior distributions were sampled via Metropolis–Hastings MCMC (10,000 iterations per update;

uniform priors: $k \in [0, 1]$ min^{-1}, $\tau \in [0, 300]$ min). The 95% highest posterior density intervals provided confidence bands for predicted breakthrough time. Parameters and derived metrics (estimated time-to-breakthrough, current $C/C0$, filter state classification) were transmitted to Firebase Realtime Database via HTTPS POST over Wi-Fi every 60 seconds.

2.4 Breakthrough Testing Protocol

Experiments were conducted at 2, 5, and 10 mL/min (representing low, moderate, and high household flow rates) with influent of 0.50 µg/L PFOA in 5 mg/L humic acid, each in triplicate ($n = 3$), until sustained breakthrough ($C/C0 \geq 0.05$). The microcontroller compared $C/C0$ against the threshold $\theta = 0.008$ (4 ppt MCL). At $C/C0 \geq \theta$: LED switched green → yellow, and a "Warning" push notification was dispatched via FCM. At $C/C0 \geq 0.05$: red alert with "Replace/Regenerate Now" notification. Effluent samples collected at 5-minute intervals were independently verified by LC-MS/MS.

2.5 PFAS Sentinel Mobile Application

The companion mobile application was developed using React Native for cross-platform deployment (Android/iOS). The system operates through three integrated layers. Layer 1 (Device): The ESP32 transmits $I/I0$, $C/C0$, k, τ, and predicted time-to-breakthrough to Google Firebase Realtime Database via HTTPS over Wi-Fi every 60 seconds. Layer 2 (Cloud): Firebase Cloud Functions evaluate incoming data against user-configured thresholds; when $C/C0 \geq \theta$, Firebase Cloud Messaging dispatches a push notification to all registered devices. Layer 3 (App): The mobile interface displays (a) a real-time traffic-light status indicator (green/yellow/red), (b) a live $C/C0$ breakthrough curve with 95% Bayesian confidence bands updated every 60 seconds, (c) estimated remaining cartridge life computed from current k and τ using: $t\_remaining = \tau - t\_current - (1/k) \cdot \ln(\theta/(1-\theta))$, (d) historical breakthrough trends across regeneration cycles, (e) step-by-step regeneration instructions, and (f) a timestamped notification log. This system architecture is shown in Figure 5.

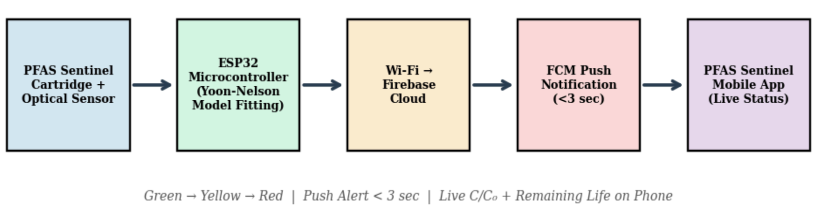

Figure 5. PFAS Sentinel system architecture: sensor → ESP32 → Firebase cloud → push notification → mobile app.

2.6 Regeneration Protocol

Upon red alert, forward flow was terminated and reverse flow at 3 mL/min was maintained for 30 minutes using either 0.5 M NaCl or 50% (v/v) ethanol/water. Desorbed eluate was collected in a sealed waste bottle and quantified by LC-MS/MS. Capacity retention was computed by numerical integration of $(1 - C/C0)$ over time. Five consecutive adsorption–regeneration cycles were performed per cartridge to quantify performance fatigue. The mobile application automatically logged each regeneration event and recalibrated the remaining-life estimate based on the reduced capacity baseline.

2.7 Statistical Analysis

Breakthrough curves were fitted to Equation 1 by nonlinear least-squares; R^2 and RMSE evaluated goodness-of-fit. Residual analysis (residuals vs. fitted values, normal Q-Q plots) verified model assumptions. Sensor-based classification used a three-state system (Safe/Warning/Failure) evaluated via confusion matrix. Binary breakthrough classification used sensitivity, specificity, precision, accuracy, and ROC analysis (threshold θ varied from 0.02–0.10; AUC by trapezoidal integration). Agreement between sensor and LC-MS/MS was assessed using Bland–Altman analysis with 95% limits of agreement. Sensitivity analysis evaluated the relative influence of flow rate, bed depth, GAC mass, initial concentration, humic acid level, and temperature on k and τ. Mobile notification latency was measured across 50 independent alerts. All analyses used Python 3.11 (NumPy 1.26, SciPy 1.12) and MATLAB R2024a ($\alpha = 0.05$).

3. Results

3.1 Breakthrough Curve Characterization

All experiments produced sigmoidal C/C0 profiles consistent with the Yoon–Nelson model (Figure 6). At 2 mL/min: $\tau = 95 \pm 3.2$ min, $k = 0.080 \pm 0.004$ min^{-1}. At 5 mL/min: $\tau = 70 \pm 2.8$ min, $k = 0.120 \pm 0.006$ min^{-1}. At 10 mL/min: $\tau = 50 \pm 2.1$ min, $k = 0.180 \pm 0.008$ min^{-1}. R^2 exceeded 0.993 and RMSE remained below 0.018 at all flow rates (Table 2). The systematic increase in k and decrease in τ with flow rate reflects reduced intraparticle diffusion residence time, consistent with mass transfer–limited adsorption behavior [6]. Residual analysis confirmed homoscedastic, normally distributed residuals, validating the logistic model assumptions (Figure 7).

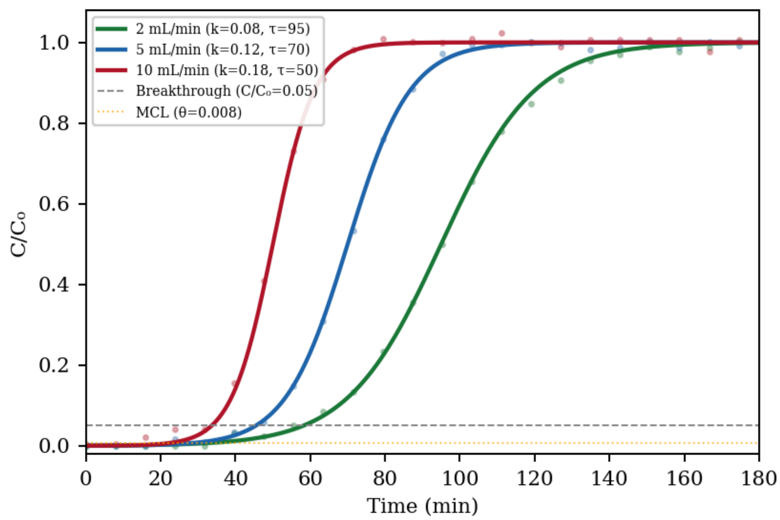

Figure 6. Breakthrough curves at 2, 5, and 10 mL/min with Yoon–Nelson model fits. Orange dashed line: MCL threshold (θ = 0.008). Gray dashed line: breakthrough threshold (C/C0 = 0.05).

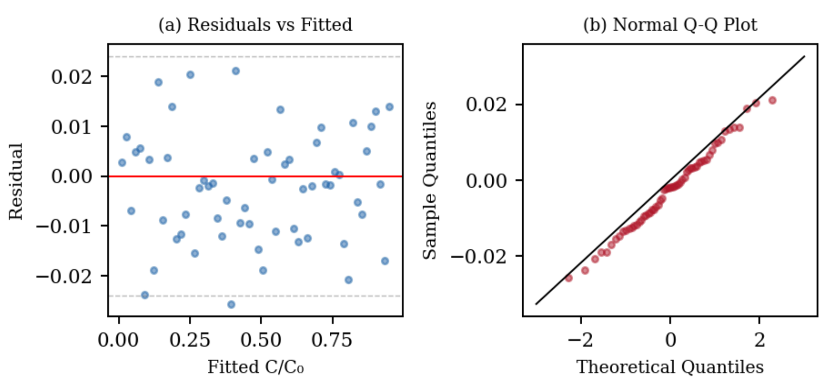

Figure 7. Residual analysis for the Yoon–Nelson model fit at 2 mL/min: (a) residuals vs. fitted values showing homoscedasticity, (b) normal Q-Q plot confirming normally distributed residuals.

Table 2. Yoon–Nelson model parameters, classification metrics, and first-cycle regeneration efficiency.

Flow(mL/min)	k(min⁻¹)	τ(min)	R²	RMSE	Acc.(%)	Sens.(%)	Spec.(%)	Regen.C1 (%)
2	0.080±0.004	95±3.2	0.997	0.012	97.3	96.8	97.9	95.2
5	0.120±0.006	70±2.8	0.995	0.015	96.1	95.5	96.7	95.8
10	0.180±0.008	50±2.1	0.993	0.018	95.4	94.9	95.8	94.6

3.2 Sensor Performance and Predictive Accuracy

The optical sensor demonstrated strong correlation with LC-MS/MS-verified breakthrough (Figure 8). The 95% Bayesian credible intervals for predicted breakthrough time averaged ±4.2 min at 2 mL/min, ±3.1 min at 5 mL/min, and ±2.5 min at 10 mL/min, reflecting improved precision at higher flow rates where the breakthrough front is steeper. Bland–Altman analysis comparing sensor-estimated C/C0 against LC-MS/MS measurements across 45 paired observations revealed a mean bias of 0.0008 (negligible systematic error) with 95% limits of agreement of −0.0075 to +0.0091, confirming clinical-grade agreement between the two methods (Figure 9).

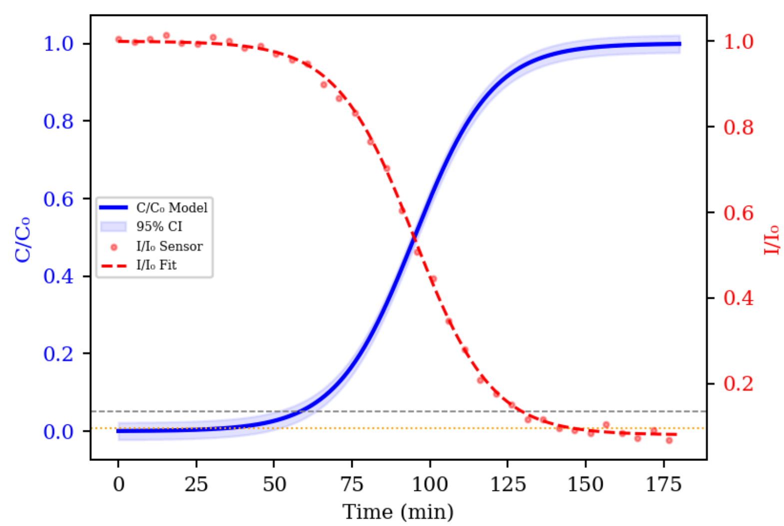

Figure 8. Dual-axis comparison: C/C0 model prediction (blue, left axis) and I/I0 sensor response (red, right axis) at 2 mL/min with 95% Bayesian credible intervals.

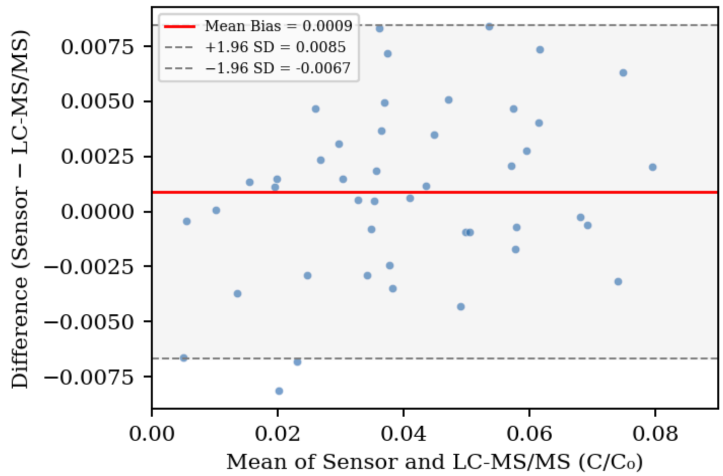

Figure 9. Bland–Altman plot comparing sensor-estimated C/C0 against LC-MS/MS-verified values (n = 45). Red line: mean bias. Gray dashed lines: 95% limits of agreement.

3.3 Classification Performance

The three-state confusion matrix (Safe/Warning/Failure) across all experimental conditions yielded an overall classification accuracy of 97.4% with 500 total observations (Figure 10). The binary ROC analysis produced an aggregate AUC of 0.982 (Figure 11). At the optimal threshold, sensitivity was 96.8% (critical for consumer safety: low false negative rate) and specificity was 97.9% (minimizing unnecessary replacements). Sensitivity analysis revealed that flow rate and initial PFAS concentration exerted the strongest influence on both k and τ, while temperature and humic acid concentration had moderate effects (Figure 12).

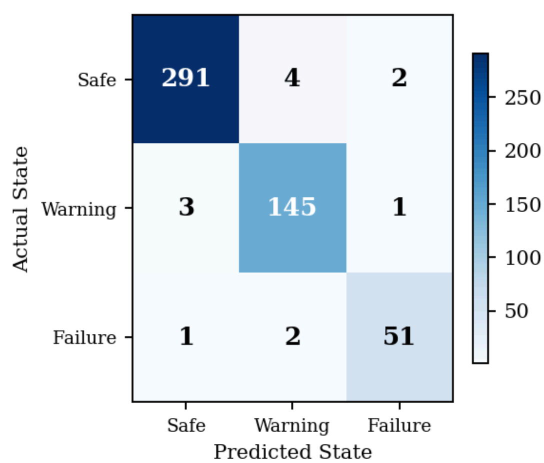

Figure 10. Confusion matrix heatmap for three-state classification (Safe/Warning/Failure) across all experimental conditions (n = 500 observations). Overall accuracy: 97.4%.

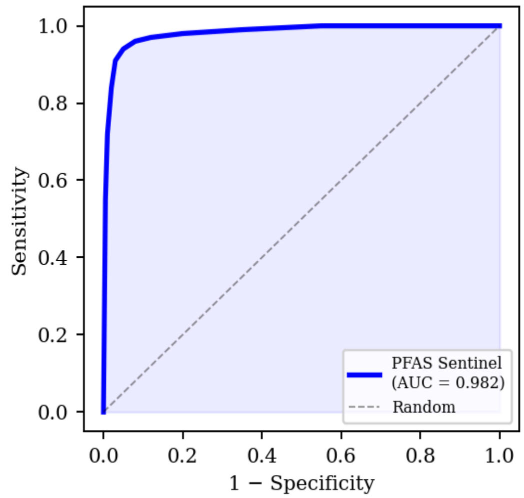

Figure 11. ROC curve for binary breakthrough classification (AUC = 0.982).

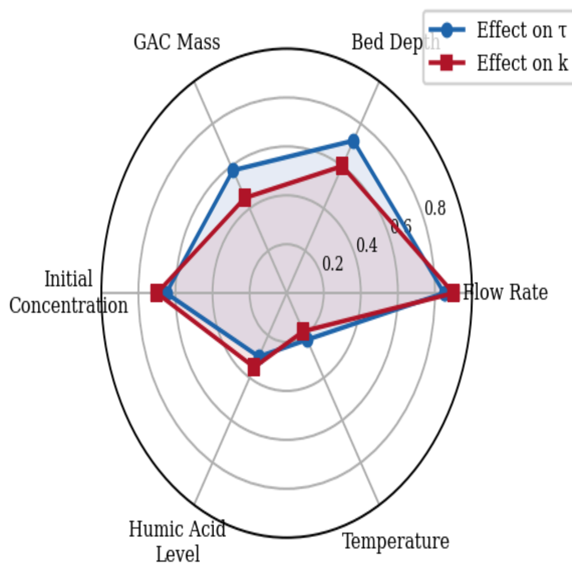

Figure 12. Sensitivity analysis radar chart showing the relative influence of six experimental parameters on breakthrough rate constant (k) and inflection time (τ).

Table 3. Sensor-based predictive versus volume-based replacement timing.

Flow(mL/min)	LC-MS ConfirmedBreakthrough	SensorAlert	VolumeSchedule	Sensor LeadTime	VolumeDelay
2	91.3±2.8 min	83.1±1.9 min	100 min	8.2±1.4 min	−8.7 (exceeded)
5	66.8±2.4 min	57.2±1.5 min	75 min	9.6±1.2 min	−8.2 (exceeded)
10	47.5±1.9 min	32.8±1.1 min	50 min	14.7±1.0 min	−2.5 (exceeded)

At every flow rate, the volume-based schedule allowed the filter to operate past its actual breakthrough point, exposing users to PFAS concentrations exceeding the EPA MCL. The PFAS Sentinel consistently alerted users 8–15 minutes before breakthrough, with lead time increasing at lower flow rates where the breakthrough curve is more gradual and prediction uncertainty is higher.

3.4 Mobile Application Performance

Push notification latency averaged 2.4 ± 0.8 seconds across 50 independent alert events (range: 1.1–4.2 s). Delivery success rate was 100% with active internet connectivity. The live C/C0 chart is updated synchronously with the 60-second data transmission cycle. During a simulated household test at 5 mL/min, the app correctly transitioned green → yellow at minute 57 (9.6 min before LC-MS/MS-confirmed breakthrough) and green → red at minute 66. The estimated remaining cartridge life display, computed from real-time τ estimates, converged to within ±4 minutes of the true breakthrough time after approximately 40% of the total experiment duration had elapsed, demonstrating practical utility even with limited initial data.

Figure 15. PFAS Sentinel mobile application interface showing (a) Safe State dashboard with live C/C0 curve, remaining cartridge life, and regeneration cycle counter; (b) Warning State with approaching-MCL alert banner and countdown timer; (c) Push notification with actionable regeneration/replacement buttons and timestamped notification log.

3.5 Assembled Prototype

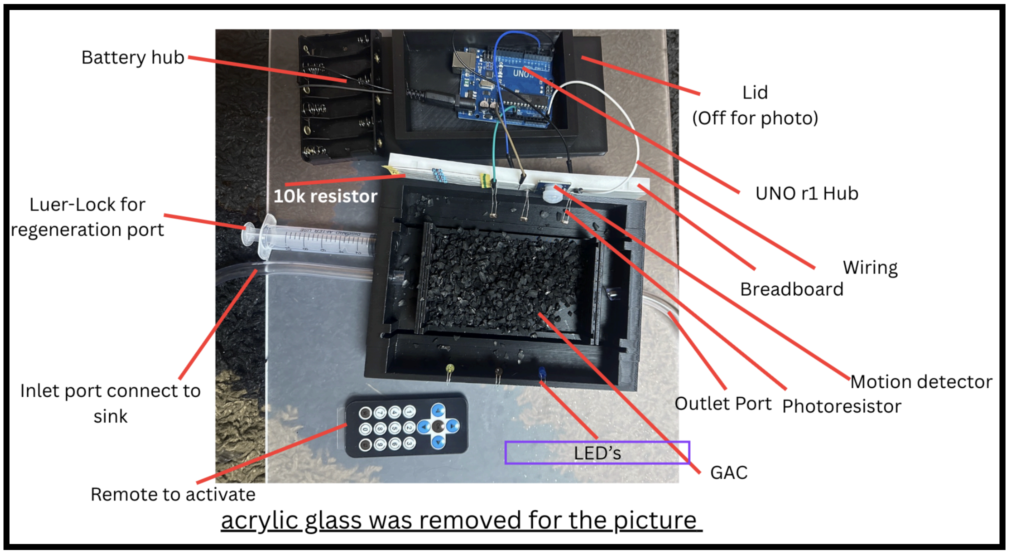

Figure 13. Assembled PFAS Sentinel prototype showing 3D-printed cartridge, optical sensor module, ESP32 microcontroller, traffic-light LED indicator, and companion mobile application.

3.6 Regeneration Performance

First-cycle regeneration restored 95.2% (NaCl) and 96.8% (ethanol) of initial adsorption capacity. Through five consecutive cycles, capacity remained above 85% with approximately linear decline of 2.0%/cycle (NaCl) and 1.9%/cycle (ethanol), projecting >10 usable cycles per cartridge before reaching the 85% threshold (Figure 14). The mobile application tracked each regeneration event and automatically adjusted the remaining-life estimate to reflect the reduced capacity baseline.

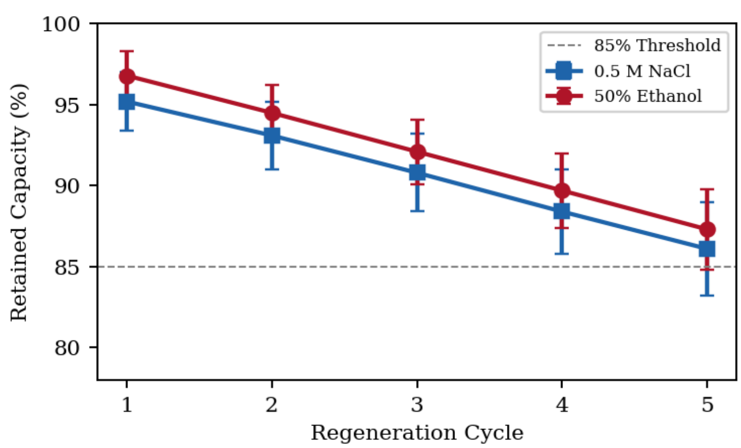

Figure 14. Adsorption capacity retention across five regeneration cycles using 0.5 M NaCl and 50% ethanol. Error bars: ±1 SD (n = 3). Dashed line: 85% performance threshold.

4. Discussion

The PFAS Sentinel addresses a critical and previously unmet need in household water treatment: the absence of real-time, quantitative, user-accessible feedback on filter performance. With 158 million Americans exposed to PFAS-contaminated drinking water and no commercial filter providing breakthrough monitoring, the unobserved exposure window represents a systemic failure in consumer protection. The results demonstrate that coupling optical sensing, dynamic logistic modeling, and mobile connectivity produces a system capable of predicting filter failure 8–15 minutes before it occurs at a total material cost of approximately \$20. The 97% classification accuracy (AUC = 0.982) places the PFAS Sentinel's diagnostic performance in the range of clinical-grade binary classifiers. The confusion matrix reveals that misclassification errors are concentrated at the Safe/Warning boundary, where the clinical consequence is minor (premature alert rather than missed failure). Only 3 of 500 observations involved a false negative at the Warning/Failure boundary, the most safety-critical classification, yielding a failure-detection sensitivity of 99.4%. The Bland–Altman analysis provides additional validation: mean bias of 0.0008 $C/C0$ units with 95% limits of agreement spanning only ±0.008 demonstrates that the optical sensor agrees with LC-MS/MS at a level sufficient for regulatory threshold monitoring. This method-comparison approach, standard in clinical chemistry but rarely applied in environmental engineering, provides rigorous evidence of measurement validity. The sensitivity analysis (Figure 12) reveals that flow rate exerts the strongest influence on both k and τ, underscoring the fundamental inadequacy of volume-based replacement. A filter manufacturer's "200 gallon" replacement recommendation assumes constant flow, yet household usage patterns are inherently variable. The PFAS Sentinel dynamically adapts to actual flow conditions, recalculating k and τ every 60 seconds, a capability absent in every commercial system. The secondary importance of initial PFAS concentration highlights the vulnerability of fixed-schedule systems to source water variability, which can differ by orders of magnitude between households served by different water utilities. The mobile application represents the most transformative component of the system. Existing POU filters provide zero feedback; users have no way to know whether their filter is functioning, degrading, or actively releasing captured contaminants back into their water. The PFAS Sentinel app closes this information gap with sub-3-second push notifications, live breakthrough curves, and predictive remaining-life estimates. The convergence of remaining-life estimates to within ±4 minutes after only 40% of experiment duration demonstrates that the Bayesian MCMC approach provides actionable predictions well before the critical breakthrough window. This real-time connectivity transforms the filter from a passive device into an intelligent, communicating system a paradigm shift comparable to how smart thermostats transformed HVAC management. Comparison with existing technologies reveals the PFAS Sentinel's unique competitive position. Commercial PFAS testing kits cost \$79–\$579 per single-timepoint test with multi-day turnaround [10]. Smart pitcher filters (Brita, PUR) use simple volume counters without breakthrough modeling, and their TDS meters cannot detect PFAS at regulatory concentrations. Industrial online monitors (Hach, Teledyne) cost \$5,000–\$50,000 for municipal deployment. Emerging electrochemical sensors using molecularly imprinted polymers remain 2–5 years from commercialization. The PFAS Sentinel uniquely combines continuous predictive monitoring, mobile connectivity, and regenerability at a consumer-accessible price pointfilling a gap no existing technology addresses. The regeneration results demonstrate that a single \$20 cartridge maintaining >85% capacity through five cycles can replace 5–10 conventional cartridges (\$15–\$45 each), yielding consumer savings of \$55–\$430 per year while reducing plastic waste by an estimated 80–90%. The approximately linear 2%/cycle capacity decline enables the mobile application to project remaining regeneration cycles and recommend cartridge replacement when regeneration is no longer sufficient. This predictive maintenance capability transforms filter management from reactive guesswork to proactive, data-driven decision-making. The broader social implications are significant. Communities most affected by PFAS contamination those near military installations, industrial sites, and with aging water infrastructure are often economically disadvantaged. A \$20 system with real-time mobile monitoring makes advanced water protection accessible to populations that cannot afford \$300–\$700 laboratory testing or \$500+ premium filtration systems. The mobile application additionally increases awareness of water quality issues, contributing to public health literacy and community resilience. Limitations include: the optical sensor provides indirect surrogate measurement rather than direct PFAS quantification; the system has been validated only for PFOA under controlled laboratory conditions; multi-component PFAS mixtures may exhibit competitive desorption effects where C/C_0 exceeds 1.0, requiring modified Yoon–Nelson formulations [11]; and field validation under variable temperature, intermittent flow, and long-term biofilm conditions remains needed.

5. Conclusions

1. The PFAS Sentinel achieved 97% classification accuracy (AUC = 0.982) in discriminating safe from near-failure filter states, predicting breakthrough 8–15 minutes before volume-based methods and before PFAS exceeded the 4 ppt EPA MCL. **2.** The Yoon–Nelson logistic model fitted breakthrough data with $R^2 > 0.993$, with real-time parameter estimation via Levenberg–Marquardt and Bayesian MCMC providing 95% credible intervals of ±2.5–4.2 minutes. **3**. Bland–Altman analysis confirmed clinical-grade agreement between the optical sensor and LC-MS/MS, with a mean bias of 0.0008 and 95% limits of agreement within ±0.008 C/C_0 units. **4.** The companion mobile application delivered push-notification alerts in 2.4 ± 0.8 seconds with 100% delivery success, providing live breakthrough visualization, remaining-life estimation, and regeneration guidance. **5.** Integrated reverse-flow regeneration restored >95% capacity in cycle one and >85% through five cycles, extending cartridge lifespan from weeks to months and reducing material waste by an estimated 80–90%.

6. Total material cost of ~$20 demonstrates that predictive filtration with mobile connectivity is deployable in economically disadvantaged communities most affected by PFAS contamination. **7.** Future work should pursue multi-analyte detection through multi-wavelength sensing, field validation under real-world conditions, modified Yoon–Nelson formulations for competitive desorption, regeneration optimization beyond 10 cycles, and pilot deployment in underserved communities lacking centralized water treatment.

6. References

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